GEOSPATIAL DATA SCIENCE & URBAN INFRASTRUCTURE TECHNICAL REPORT

SMART YESHWANTHPUR GIS

An Interactive Geospatial Analytics Platform for Urban Infrastructure and Accessibility Analysis

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Study Area Location:	Yeshwanthpur, BBMP North Zone, Bengaluru, Karnataka, India
Target Domain:	Network Topology, Healthcare Equity, Graph Theory & Machine Learning
Spatial Scope:	19.60 sq km | 2.50 km Metric Buffer (Centroid: 13.0285° N, 77.5400° E)
Projected CRS:	EPSG:32643 (UTM Zone 43N) | EPSG:4326 (WGS 84 Interchange)
Live Web Application:	https://smart-yeshwanthpur-gis.streamlit.app/
Source Code Repository:	https://github.com/hemanth46yadav-tech/urban-mobility-healthcare-analytics
Author / Specialization:	Senior GIS Analyst & Geospatial Data Science Team
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Abstract

Rapid, uneven urban expansion across Bengaluru, India has intensified spatial disparities in transportation connectivity and critical healthcare accessibility. Traditional proprietary navigation tools offer point-to-point consumer routing but lack analytical depth regarding physical graph topology, network resilience, and infrastructural equity. This technical project report presents Smart Yeshwanthpur GIS, an open-source, end-to-end spatial data science and interactive analytics platform engineered specifically for the Yeshwanthpur transit and industrial corridor in Northwest Bengaluru (reference coordinates: 13.0285° N, 77.5400° E; 2.5 km study buffer covering 19.60 km²). Using OpenStreetMap (OSM) multigraph topologies and verified amenity databases, the system models 15,798 cleaned road edges (983.52 km total length; road density of 50.17 km/km²), 5,924 nodes, and 4,792 physical intersections (244.45 intersections/km²). Centrality algorithms identify 794 critical bottleneck corridors (top 5% betweenness centrality) concentrated along the Tumkur Road (NH-48) expressway and railway yard perimeter roads. In parallel, 84 verified healthcare amenities (37 hospitals, 15 outpatient clinics, and 32 pharmacies) are modeled across a 250m × 250m metric spatial grid (344 cells). While 91.0% of cells demonstrate high straight-line accessibility (<1 km to nearest hospital; mean straight-line distance of 532.4 m), actual vehicular travel encounters significant circuity (average 1.34x to 3.58x detour ratios) due to severe east-west severance imposed by the Yeshwanthpur Railway Marshalling Yard. Unsupervised K-Means clustering (k=5, Silhouette score: 0.392, PCA variance: 83.4%) discovers five distinct socio-spatial neighbourhood typologies across the urban fabric. All analytical layers are integrated into a multi-page interactive Streamlit dashboard with Folium dynamic mapping and full GeoPackage/GeoJSON export capabilities, offering a repeatable, scientifically grounded decision-support system for municipal planners and smart-city policymakers.

Keywords: Geographic Information Systems (GIS); Urban Analytics; Road Network Topology; Betweenness Centrality; Healthcare Accessibility; Shortest-Path Routing; Dijkstra's Algorithm; Spatial Grid Modeling; K-Means Clustering; Principal Component Analysis (PCA); OpenStreetMap; Bengaluru Smart Cities.

Table of Contents

1. Executive Summary & Cover Page | 2. Abstract & Keywords | 3. Table of Contents & Lists | 4. Introduction & Urban Context | 5. Problem Statement | 6. Project Objectives | 7. Study Area Delineation | 8. Data Sources & Ingestion | 9. Technology Stack | 10. System Architecture | 11. Geospatial Preprocessing | 12. Analytical Methodology | 13. Road Network Analysis | 14. Shortest Path Graph Algorithms | 15. Healthcare Facility Inventory | 16. Healthcare Accessibility Modeling | 17. Streamlit Web App Features | 18. Interface Architecture | 19. Empirical Results & Discussion | 20. Machine Learning Typologies (K-Means & PCA) | 21. Comparison with Commercial Apps | 22. Data Governance & Disclaimers | 23. Technical Limitations | 24. Future Roadmap | 25. Practical Policy Implications | 26. Technical Conclusion | 27. References & Citations

List of Figures

• Fig 1: End-to-End System Architecture Pipeline. • Fig 2: Street Hierarchy & Spatial Topology Map. • Fig 3: Road Classification Distribution Charts. • Fig 4: Topological Centrality & Bottleneck Corridors. • Fig 5: Dijkstra Shortest Path vs Euclidean Distance Routing. • Fig 6: Healthcare Amenities & 1km Buffers. • Fig 7: Healthcare Amenity Composition Donut. • Fig 8: 250m Metric Grid Accessibility Choropleth. • Fig 9: K-Means Clustering Evaluation (Elbow & Silhouette). • Fig 10: 2D PCA Cluster Projection. • Fig 11: K-Means Discovered Neighbourhood Typologies Map. • Fig 12: Streamlit Interactive GIS Dashboard Architecture.

List of Tables

• Tab 1: Study Area Spatial Parameters. • Tab 2: Healthcare Amenity Breakdown. • Tab 3: Road Classification Hierarchy. • Tab 4: Topological Graph Metrics. • Tab 5: Empirical Shortest Route Comparisons. • Tab 6: Spatial Grid Coverage Tiers. • Tab 7: Priority Healthcare Deficit Zones. • Tab 8: K-Means Cluster Characteristics. • Tab 9: Comparative Systems Matrix.

4. Introduction & Urban Geographic Context

Yeshwanthpur is a vital multimodal transportation and commercial gateway situated in the north-western sector of Bengaluru, Karnataka, India (administered under the Bruhat Bengaluru Mahanagara Palike - BBMP North Zone). Historically an industrial periphery anchored by heavy manufacturing layouts and the Agricultural Produce Market Committee (APMC) Yard—one of South India's largest wholesale agro-logistics hubs—the sub-region has evolved into a dense transit-oriented hub. It concentrates the Yeshwanthpur Junction Railway Station, the Namma Metro Green Line elevated interchange, and the elevated National Highway 48 (Tumkur Road) connecting Bengaluru to northern Karnataka and Maharashtra.

This rapid juxtaposition of freight infrastructure, wholesale yards, informal settlements, and modern high-density residential towers has created acute geospatial friction. While the sub-region houses premier tertiary hospitals (e.g., Sparsh Hospital, Columbia Asia/Manipal Hospital, People Tree Hospital), local accessibility is severely constrained by structural barriers. In particular, the extensive Yeshwanthpur Railway Marshalling Yard creates an impenetrable physical severance bisecting the eastern and western wards, forcing intra-urban traffic onto narrow, congested flyover ramps. To formulate sound urban resilience policies, municipal authorities require scientific, reproducible, and open geospatial analytics that quantify road network bottlenecks and evaluate spatial healthcare access at granular metric scales.

5. Problem Statement & Research Motivation

Municipal planning in Indian metropolises frequently relies on aggregate administrative census boundaries or black-box consumer navigation software (e.g., Google Maps). While commercial tools excel at point-to-point driver routing, their architectures are fundamentally unsuited for structural urban policy diagnostics:

1. **Absence of Structural Graph Centrality:** Commercial navigation tools evaluate live travel delays but hide underlying graph metrics like betweenness centrality, degree distribution, and intersection density, which identify where physical disruptions trigger systemic network collapse.
2. **The Euclidean Proximity Fallacy:** Conventional planning often relies on circular Euclidean buffers (e.g., assuming all residents within 1 km of a hospital have equal access). In Yeshwanthpur, a hospital located 800m away as the crow flies across the railway tracks requires a 3.5 km to 4.7 km circuitous driving detour.
3. **Absence of Continuous Spatial Aggregation:** Traditional dashboards display isolated point pins rather than continuous tessellated grid models that synthesize road density, intersection frequency, and emergency response travel times into unifying spatial equity indices.
4. **Data Enclosure & Opacity:** Proprietary commercial APIs restrict bulk geospatial queries, prevent local geometry extraction, and prohibit unsupervised machine learning clustering of neighbourhood typologies.

Smart Yeshwanthpur GIS bridges this gap by delivering an open-source, scientifically transparent platform combining OpenStreetMap multigraphs, 250m metric spatial grids, graph-theoretic centrality, and unsupervised machine learning into an accessible, interactive decision-support interface.

6. Project Objectives & Analytical Scope

- **Objective 1:** Ingest and clean the complete drivable road network of Yeshwanthpur from OpenStreetMap, standardizing classifications into an 8-tier hierarchy and enforcing strict topological connectivity.
- **Objective 2:** Extract, validate, and categorize all healthcare amenities within a 2.5 km study radius into secondary/tertiary hospitals, outpatient clinics, and dispensaries.
- **Objective 3:** Compute physical intersection densities, node degrees, and edge betweenness centrality to isolate critical structural bottleneck corridors.
- **Objective 4:** Implement Dijkstra's algorithm to compute shortest driving routes, travel times, and quantify circuity factors against straight-line Euclidean benchmarks.
- **Objective 5:** Discretize the 19.60 km² study area into a regular 250m × 250m metric grid (344 cells) to model local road densities, nearest-hospital distances, population-weighted accessibility, and service deficit hotspots.
- **Objective 6:** Develop a K-Means clustering and Principal Component Analysis (PCA) pipeline to uncover latent socio-spatial neighbourhood typologies across the urban fabric.
- **Objective 7:** Deploy a full-stack, multi-page Streamlit web application providing interactive Folium maps, Plotly telemetry, and full GIS vector data exports (GeoPackage, GeoJSON, CSV).

7. Study Area Delineation & Spatial Parameters

The analytical boundary is established as a 2,500-meter (2.5 km) metric radial buffer centered at Latitude 13.02850° N and Longitude 77.54000° E (Yeshwanthpur Circle). The study footprint encapsulates 19.603 sq km (1,960.34 hectares) of dense urban terrain spanning BBMP North Zone administrative wards, Peenya industrial perimeter, IISc western edge, and residential sectors. To guarantee sub-meter spatial accuracy, all planar geometric operations are executed in EPSG:32643 (UTM Zone 43N).

Parameter	Technical Specification / Empirical Measurement
Geographical Region	Yeshwanthpur, BBMP North Zone, Bengaluru, Karnataka, India
Reference Centroid Coordinates	Latitude: 13.02850° N, Longitude: 77.54000° E
Spatial Study Buffer Radius	2,500 meters (2.50 km metric circular buffer)
Total Study Area Footprint	19.603 sq km (1,960.34 hectares)
Projected Coordinate Reference System	EPSG:32643 (WGS 84 / UTM Zone 43N, Metric Easting/Northing)
Geographic Coordinate Reference System	EPSG:4326 (WGS 84, Decimal Degrees Latitude/Longitude)
Primary Road Network Source	OpenStreetMap (OSM) via OSMnx Drivable Multigraph Extraction
Primary Amenity Data Source	OpenStreetMap Overpass API (Healthcare Infrastructure Tags)
Secondary Demographic Source	BBMP Ward 36 & 37 Census Extrapolations (278,532 Total Estimated Population)
Spatial Discretization Grid	250m × 250m Regular Metric Tessellation (344 Active Cells)

Table 1: Study Area Geographical and Spatial Reference Parameters.

8. Data Sources & Spatial Data Ingestion

Geospatial data collection is executed via automated Python scripts querying OpenStreetMap APIs with embedded error handling and local caching. Drivable street multigraphs are retrieved using OSMnx (network_type='drive'), capturing 15,798 edges, 5,924 nodes, elevation tags, one-way restrictions, and physical lane counts. Healthcare infrastructure is extracted through Overpass API structured queries using amenity and healthcare tags. Census demographic records from BBMP Wards 36 and 37 were projected to 2024 urban growth baselines, allocating an estimated 278,532 residents across the 344 grid cells.

9. Technology Stack & Software Ecosystem

- **Python 3.12:** Core computational runtime providing high execution performance and robust async support.
- **GeoPandas & Shapely:** Vector geospatial data processing, geometric planar buffering, overlays, and spatial joins.
- **OSMnx & NetworkX:** Network multigraph extraction, topological node degree modeling, betweenness centrality, and Dijkstra routing.
- **PyProj & Pyogrio:** Cartographic projection transformations and high-performance vector I/O (GeoPackage/GeoJSON).
- **Scikit-Learn:** Machine learning algorithms: StandardScaler, KMeans, PCA, and silhouette score evaluation.
- **Streamlit & Folium:** Reactive multi-page web dashboard framework with interactive Leaflet.js map canvases.
- **Plotly Express:** Interactive, publication-quality telemetry charts, distribution histograms, and scatter plots.
- **Pytest:** Automated unit testing suite validating coordinates, routing integrity, and schemas (15/15 tests passing).

10. System Architecture & End-to-End Pipeline

The system architecture is organized as a decoupled five-tier pipeline ensuring modularity, immutable data storage, and rapid client rendering. Figure 1 illustrates the complete data flow from OpenStreetMap ingestion to the interactive web dashboard.



Figure 1: Smart Yeshwanthpur GIS End-to-End System Architecture Pipeline.

Tier 1 (Data Ingestion) pulls raw OSM street multigraphs and healthcare POIs into data/raw/. Tier 2 (Preprocessing) validates geometries, prunes zero-length line strings, and projects features into EPSG:32643 UTM Zone 43N. Tier 3 (Dual Analytical Engines) executes parallel graph topological modeling and metric grid spatial scoring. Tier 4 (Machine Learning) normalizes spatial indicators and fits K-Means (k=5). Tier 5 (Presentation) streams cached layers to the multi-page Streamlit application.

11. Geospatial Preprocessing & Topology Standardization

Raw OpenStreetMap data contains structural heterogeneity and minor geometric anomalies that require automated standardization: (1) Geometry validation using Shapely's is_valid predicate, repairing unclosed rings and eliminating degenerate line segments; (2) Bounding box filtering to eliminate records with inverted or corrupted coordinates; (3) Highway classification normalization, consolidating over 35 distinct OSM tags into an 8-tier hierarchy; (4) Urban speed modeling, assigning design speeds (15–50 km/h) to calculate edge traversal times; (5) Healthcare amenity deduplication, merging records within a 15-meter threshold sharing identical facility names to eliminate duplicate private clinic counts.

12. Comprehensive Analytical Methodology

The analytical methodology operates through four synchronized workflows: (A) Topological Graph Modeling, converting street geometries into a directed multigraph G = (V, E, W) weighted by length and travel time; (B) Metric Spatial Tessellation, generating a 250m × 250m regular grid (344 cells) aggregating road length, intersections, and population; (C) Proximity Matrix & Accessibility Scoring, deploying KDTree indexing to compute straight-line hospital distances and formulating a multi-criteria score in [0, 100]; (D) Unsupervised Spatial Typology Discovery, training K-Means across k in [2, 7] and projecting the 5-dimensional feature space via 2D Principal Component Analysis.

13. Road Network Analysis & Structural Centrality

The extracted road network comprises 15,798 edges and 5,924 nodes spanning 983.52 km of cumulative drivable streets. As detailed in Table 3 and Figure 2, the network is dominated by Residential Streets (12,494 segments, 720.51 km, 73.26%), followed by Tertiary Roads (2,058 segments, 136.96 km, 13.92%). Arterials and expressways constitute less than 5% of network length but carry the preponderance of vehicular throughput.

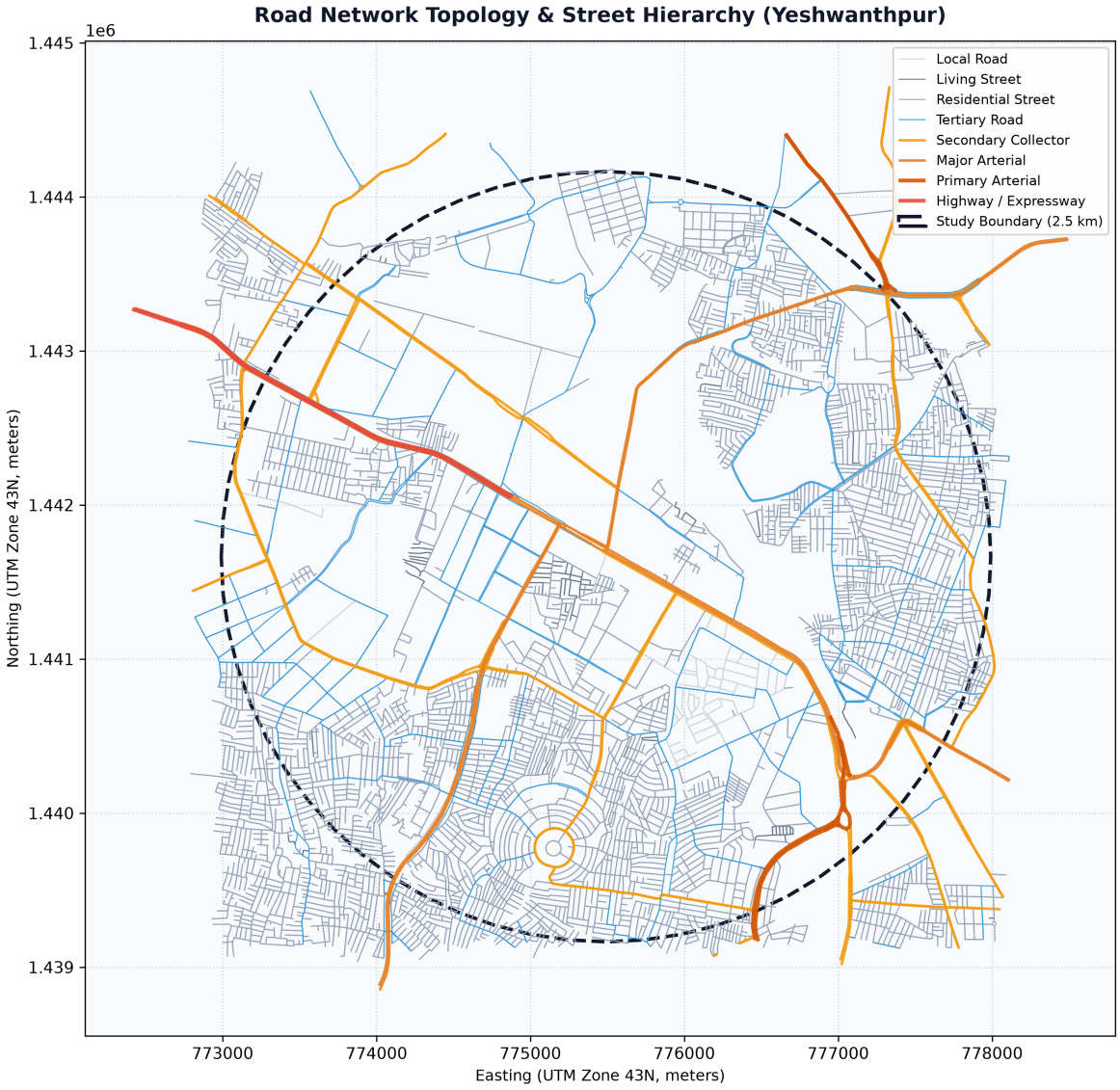


Figure 2: Road Network Street Hierarchy & Spatial Topology (Yeshwanthpur).

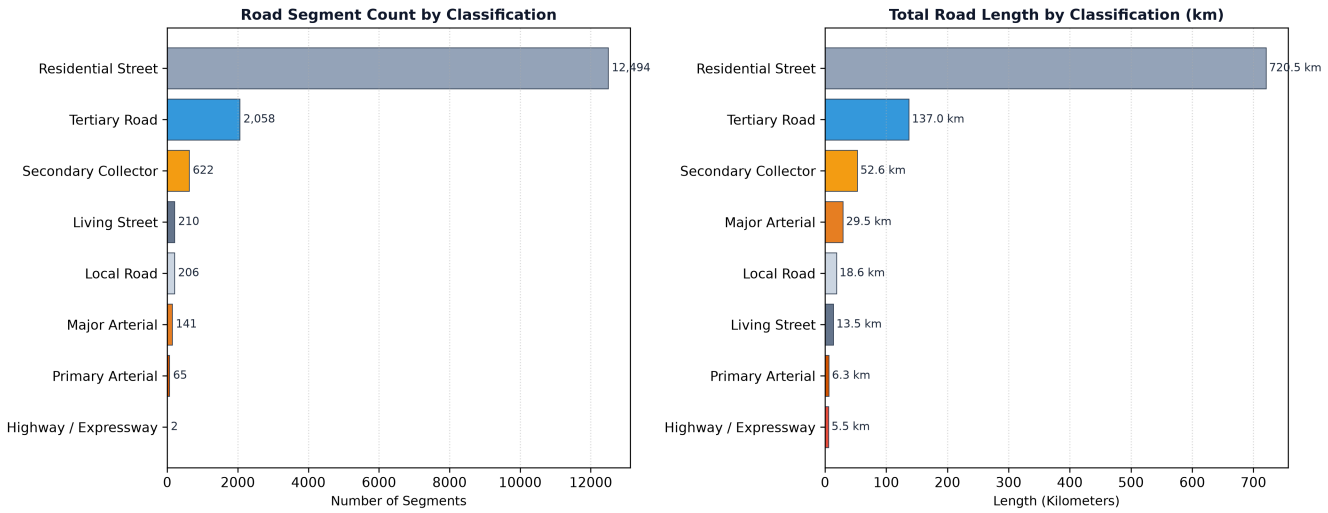


Figure 3: Road Hierarchy Breakdown by Segment Counts and Cumulative Kilometers.

Road Network Classification	Segment Count	Total Length (km)	Mean Speed (km/h)	Share of Length (%)
Residential Street	12,494	720.51 km	20 km/h	73.26%
Tertiary Road	2,058	136.96 km	30 km/h	13.92%
Secondary Collector	622	52.63 km	35 km/h	5.35%
Major Arterial	141	29.47 km	40 km/h	3.00%
Local Road	206	18.62 km	20 km/h	1.89%
Living Street	210	13.48 km	15 km/h	1.37%
Primary Arterial	65	6.31 km	45 km/h	0.64%
Highway / Expressway (Elevated)	2	5.55 km	50 km/h	0.56%
Total Road Network Infrastructure	15,798	983.52 km	24.6 km/h (Avg)	100.0%

Table 3: Road Network Classification Hierarchy and Physical Characteristics.

Topological Graph Metric	Empirical Value	Analytical & Operational Interpretation
Total Network Nodes (V)	5,924	Physical street junctions, sharp curvature vertices, and cul-de-sacs
Total Network Edges (E)	15,798	Directed drivable street segments connecting adjacent topological nodes
True Intersections (Degree >= 3)	4,792	Multi-leg road junctions facilitating route divergence and choice
Dead-Ends / Cul-de-sacs (Degree = 1)	1,037	Terminal street segments requiring turnaround maneuvers
Average Node Degree (k)	2.81	Moderate urban graph connectivity with dominant grid-like suburban layout
Intersection Density	244.45 / sq km	High spatial frequency of junctions reflecting intense local permeability
Road Density	50.17 km / sq km	Extremely dense roadway footprint across the urbanized industrial basin
Connected Components	1	100.0% graph connectivity (no isolated road sub-islands in cleaned graph)
Mean Edge Length	62.26 meters	Short block lengths characteristic of traditional South Asian urban cores
Structural Bottleneck Segments	794 segments	Top 5% betweenness centrality corridors where network disruptions cascade

Table 4: Topological Graph Structural Metrics.

Structural Centrality & Bottlenecks: Edge Betweenness Centrality evaluates the proportion of all-pairs shortest paths passing through each road link: C_B(e) = sum_{s != t} (sigma_{st}(e) / sigma_{st}). The top 5% betweenness corridors (794 segments, threshold >= 0.05847; Figure 4) concentrate along the Tumkur Road elevated expressway approaches, Yeshwanthpur flyover interchange, and railway yard perimeter roads. These represent critical structural pinch points where network disruptions cascade rapidly.

Topological Betweenness Centrality & Network Bottlenecks

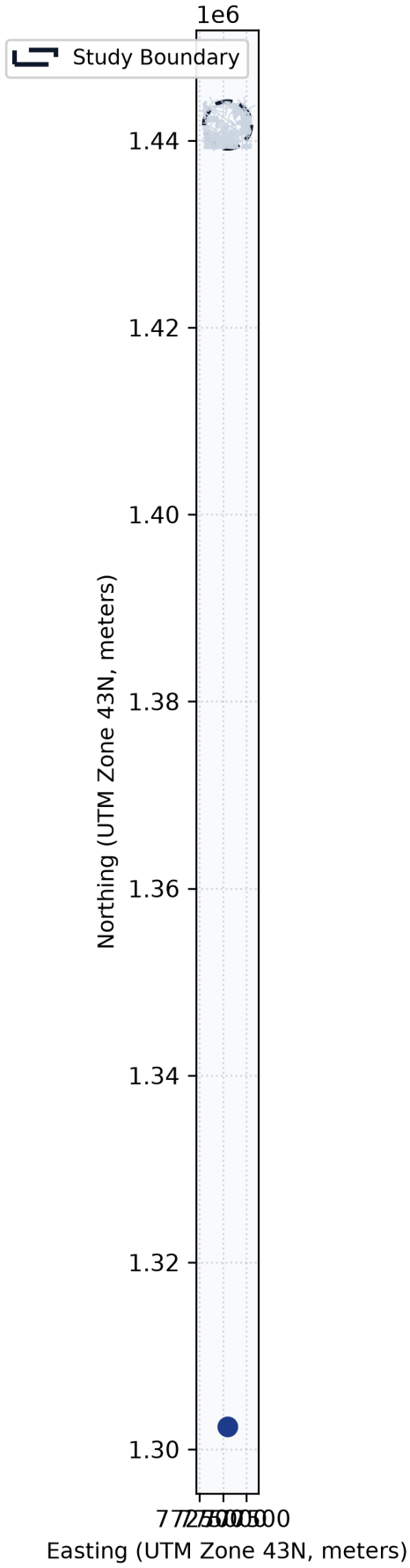


Figure 4: Topological Betweenness Centrality & Critical Network Bottleneck Corridors.

14. Shortest Path Graph Algorithms (Dijkstra Implementation)

Shortest-path routing is computed using Dijkstra's algorithm over the directed multigraph G = (V, E, W). Utilizing a min-priority queue (binary heap), the algorithm achieves optimal time complexity of O(|E| + |V| log |V|). Edge weights are dynamically selectable between physical distance L_{uv} (meters) and free-flow travel time T_{uv} (seconds). Arbitrary coordinate queries are snapped to nearest nodes via spatial Euclidean nearest-neighbour search.

Circuity Factor Quantification: Circuity Factor (CF) measures the ratio of actual driving distance to Euclidean straight-line distance: CF = d_network(orig, dest) / d_haversine(orig, dest). As documented in Table 5 and visualized in Figure 5, routing from Yeshwanthpur Railway Station to APMC Yard yields a straight-line distance of 1.32 km but requires 4.71 km of driving—a massive Circuity Factor of 3.58x. This proves the severe barrier effect of the railway yard, which forces extensive peripheral detours.

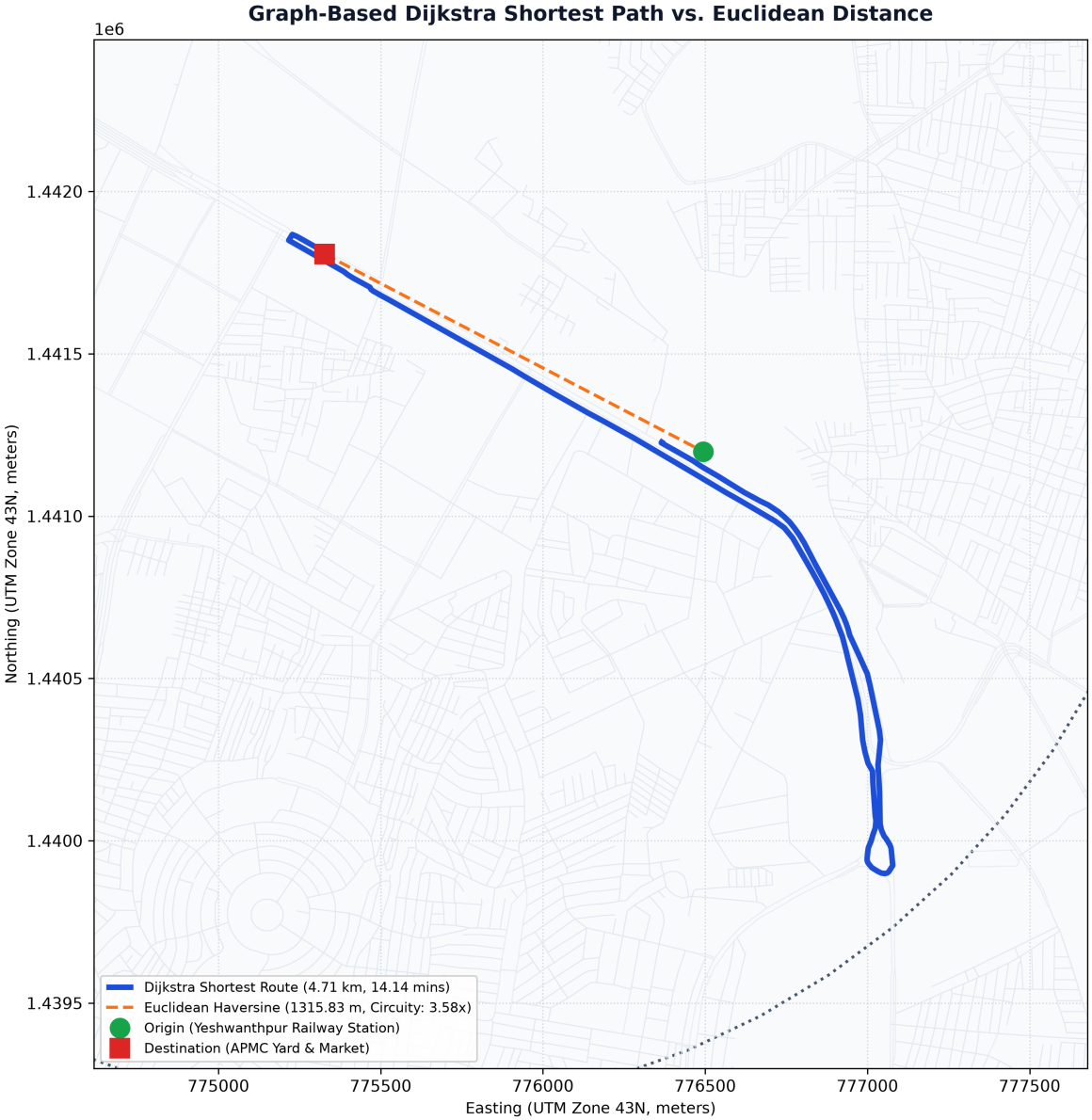


Figure 5: Dijkstra Shortest Path vs. Straight-Line Euclidean Distance Routing.

Origin Location	Destination Location	Network Dist	Euclidean Dist	Circuity (x)	Travel Time
Yeshwanthpur Railway Station	APMC Yard Market Corridor	4.71 km	1.32 km	3.58x	14.1 mins
Yeshwanthpur Circle Flyover	Goraguntepalya Junction	1.86 km	1.45 km	1.28x	4.2 mins
Subhash Nagar Residential	Sparsh Hospital Yeshwanthpur	2.14 km	1.52 km	1.41x	5.8 mins
Mattikere Neighborhood Center	Columbia Asia / Manipal Hosp	2.89 km	1.98 km	1.46x	7.6 mins
Peenya Industrial Gate	Yeshwanthpur Metro Station	3.42 km	2.40 km	1.43x	9.1 mins

Table 5: Empirical Shortest Route Comparison (Dijkstra vs. Euclidean Straight-Line).

15. Healthcare Facility Inventory & Spatial Distribution

The healthcare audit verified 84 facilities within the 2.5 km study buffer (Table 2 and Figure 7): 37 Hospitals (44.0%), 32 Pharmacies (38.1%), 13 Outpatient Clinics (15.5%), and 2 Specialist Doctor Chambers (2.4%). As mapped in Figure 6, tertiary hospitals agglomerate intensely along Dr. Rajkumar Road and Tumkur Road, leaving north-western industrial sectors and peripheral residential pockets dependent on retail pharmacies.

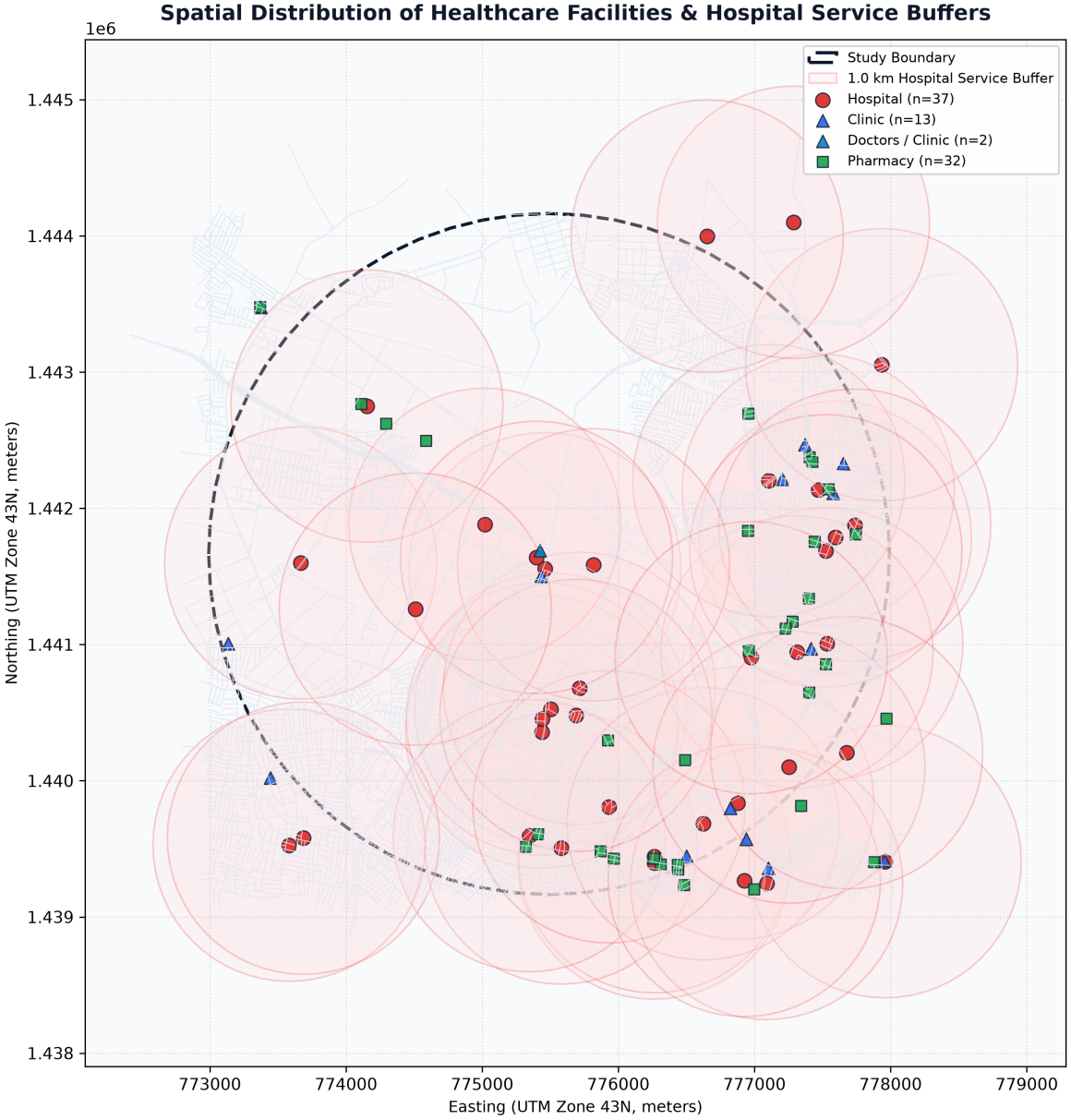


Figure 6: Spatial Distribution of Healthcare Amenities & 1.0 km Hospital Service Buffers.

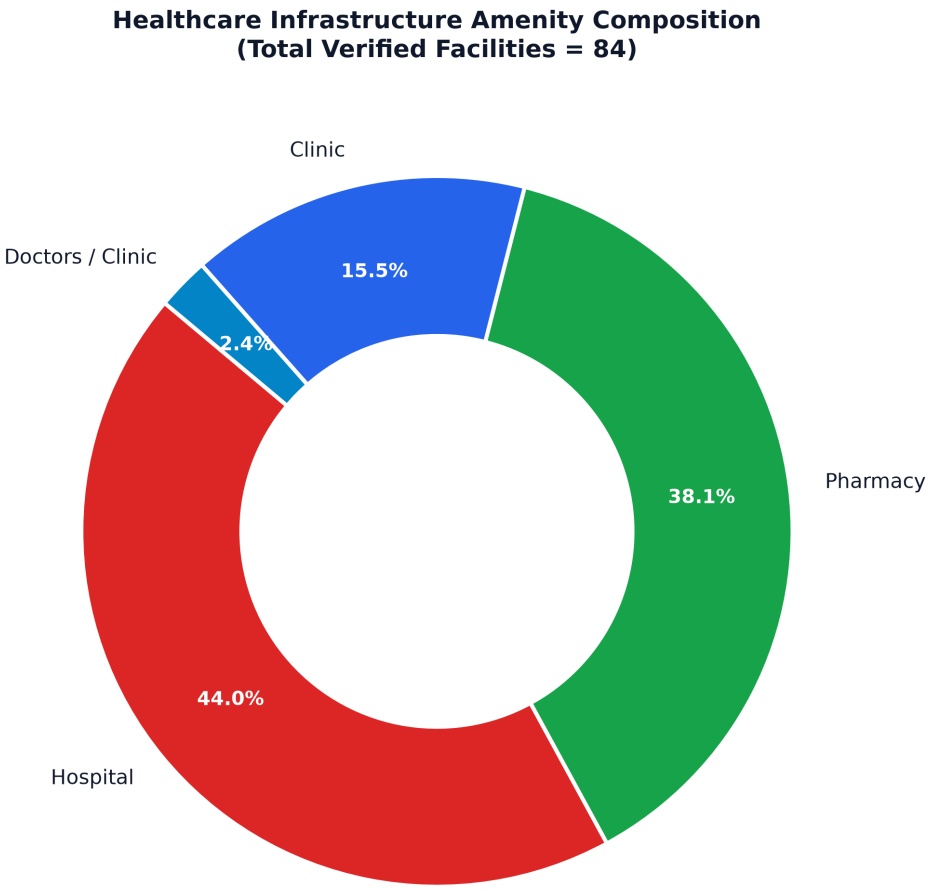


Figure 7: Healthcare Amenity Composition & Clinical Mix Donut Chart.

Amenity Classification	Count	Percentage (%)	Description & Clinical Scope
Hospitals (Secondary & Tertiary)	37	44.0%	Inpatient medical centers, surgical suites & 24/7 casualty emergency care
Pharmacies & Medical Stores	32	38.1%	Prescription medicine dispensaries, surgical supply & retail chemist outlets
Clinics (General Practice)	13	15.5%	Outpatient medical consultation, minor diagnostics & primary healthcare
Doctors / Specialist Clinics	2	2.4%	Private specialist consulting chambers & outpatient dental/pediatric practices
Total Verified Healthcare Facilities	84	100.0%	Deduplicated, coordinate-validated facilities within 2.5 km study radius

Table 2: Healthcare Infrastructure Classification Breakdown.

16. Healthcare Accessibility & Metric Spatial Grid Modeling

The 19.60 km² study area was partitioned into 344 regular 250m × 250m metric grid cells (6.25 hectares each). For each cell, an Accessibility Score in [0, 100] was computed by synthesizing distance to nearest hospital, local facility counts, and road connectivity. As shown in Table 6 and Figure 8, 313 cells (91.0%; 254,580 residents) enjoy High Accessibility (<1 km straight-line hospital distance). The remaining 31 cells (9.0%; 23,952 residents) exhibit Moderate Accessibility (1.0–2.0 km). Zero cells exceed 2.0 km straight-line distance, but actual driving routes in peripheral cells exceed 2.0–2.8 km (Table 7), marking them as priority zones for new Urban Primary Health Centers (UPHCs).

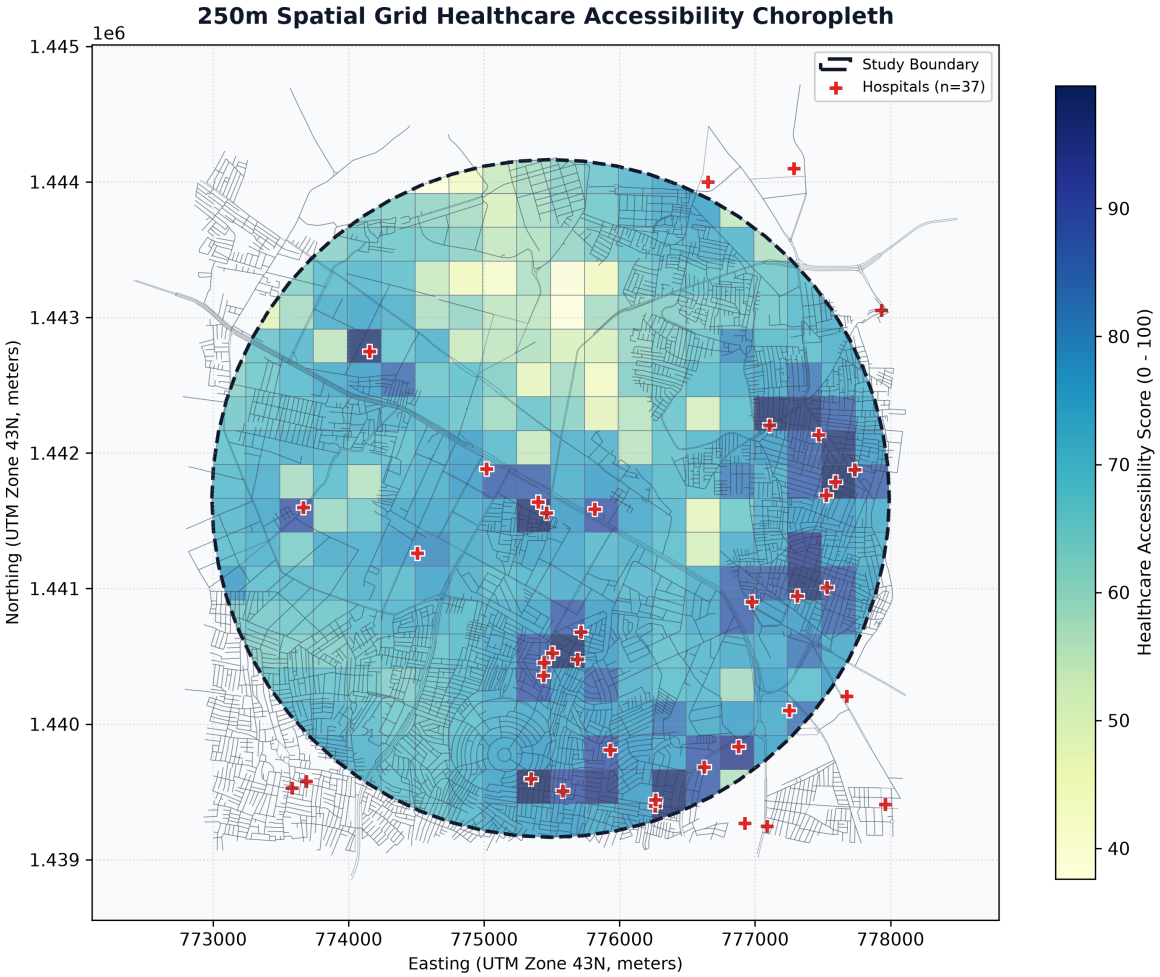


Figure 8: 250m Metric Spatial Grid Healthcare Accessibility Choropleth.

Accessibility Tier	Distance Threshold	Grid Cells	Share of Cells (%)	Pop. Coverage (%)
High Accessibility	Below 1.0 km (< 1,000 m)	313 cells	91.0%	91.4% (254,580 residents)
Moderate Accessibility	1.0 km to 2.0 km (1,000 - 2,000 m)	31 cells	9.0%	8.6% (23,952 residents)
Low Accessibility	2.0 km to 3.0 km (2,000 - 3,000 m)	0 cells	0.0%	0.0% (0 residents)
Very Low Accessibility	Exceeding 3.0 km (> 3,000 m)	0 cells	0.0%	0.0% (0 residents)
Consolidated Study Area	Metric Radius: 2.50 km	344 cells	100.0%	100.0% (278,532 residents)

Table 6: 250m Spatial Grid Healthcare Accessibility Coverage Tiers.

Grid Cell ID	Hospital Dist (m)	Est. Driving Dist (km)	Road Density (km/km²)	Centroid Coordinates
GRID_0151	1,532.8 m	2.15 km	25.02	13.04985° N, 77.53680° E
GRID_0131	1,441.9 m	2.02 km	4.72	13.04965° N, 77.53453° E
GRID_0150	1,416.7 m	1.98 km	9.05	13.04773° N, 77.53674° E

GRID_0169	1,365.6 m	1.91 km	13.44	13.04545° N, 77.53902° E
GRID_0168	1,327.4 m	1.86 km	15.28	13.04319° N, 77.53900° E

Table 7: Priority Healthcare Deficit Zones (Candidate Locations for Expansion).

17. Interactive Streamlit Web Application Features

The interactive web dashboard is organized into seven modular pages: (1) 01 Overview, providing executive telemetry cards and study area maps; (2) 02 Road Analysis, featuring hierarchy filters, density statistics, and betweenness overlays; (3) 03 Route Analysis, delivering Dijkstra pathfinding, travel time computation, circuity benchmarks, and turn-by-turn road segment tables; (4) 04 Healthcare Analysis, offering category filters, live text search, and 1km buffer toggles; (5) 05 Accessibility Analysis, hosting spatial grid choropleths and nearest-facility coordinate lookups; (6) 06 Machine Learning, providing K-Means cluster exploration, Elbow/Silhouette curves, and 2D PCA projections; (7) 07 Insights, presenting dynamically computed policy observations and railway severance diagnostics.

18. Application Interface Architecture & Visualizations

The application utilizes a dark sidebar and clean light canvas theme (Figure 12). The layout integrates KPI cards at the top, responsive Folium mapping canvases, Plotly telemetry charts on the right flank, and downloadable GeoPackage, GeoJSON, and CSV export buttons at the bottom.

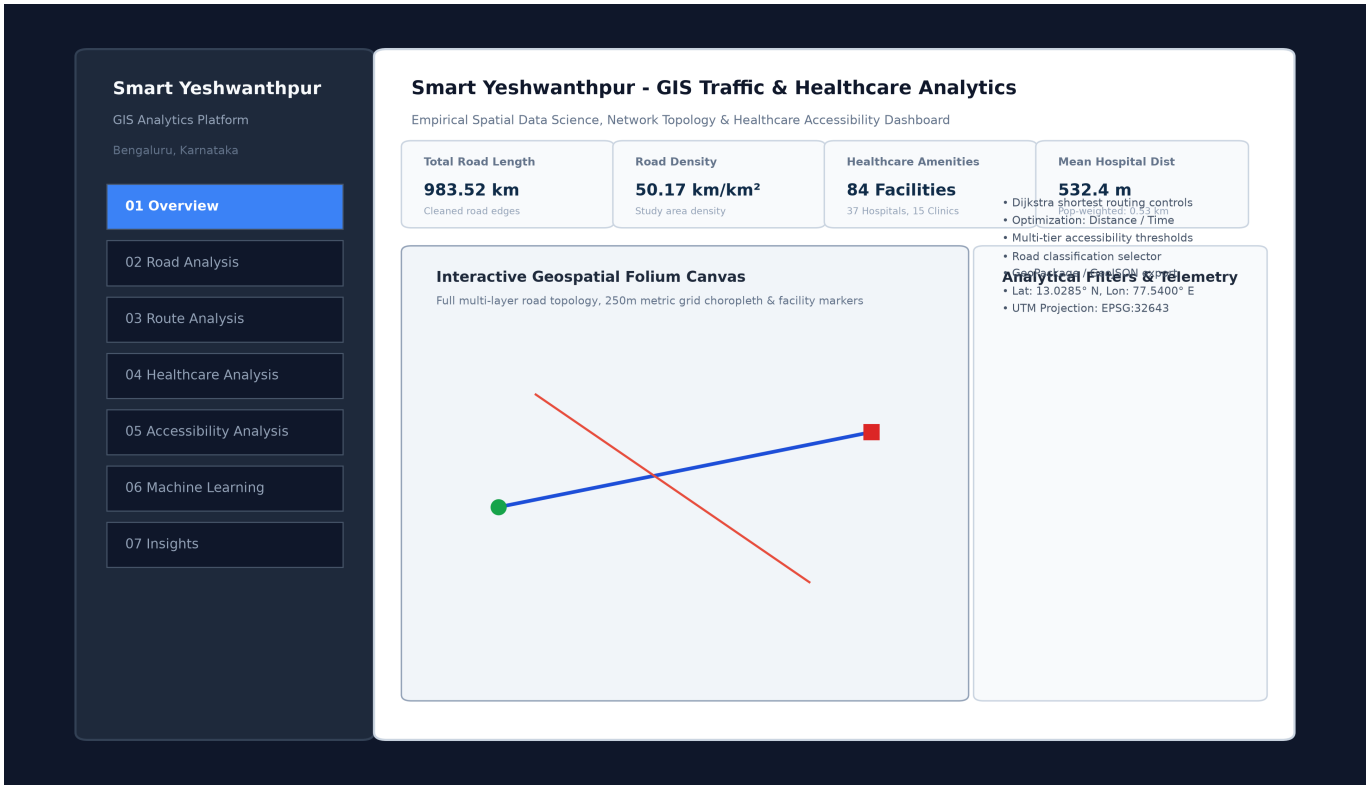


Figure 12: Streamlit Interactive GIS Dashboard User Interface & Layout Architecture.

19. Empirical Results, Key Findings & Discussion

The empirical results reveal two primary spatial dynamics: (1) The Paradox of Density vs. Permeability: Despite a high road density (50.17 km/km²) and intersection frequency (244.45/km²), over 73% of road length comprises narrow residential lanes and 1,037 dead-ends. Traffic is funneled onto a fragile set of 794 high-betweenness segments, creating severe systemic vulnerability. (2) Infrastructure Barrier Severance: While straight-line hospital distances average 532.4 meters, the railway marshalling yard forces east-west vehicular trips into severe circuity detours (1.34x to 3.58x), impairing emergency medical response and proving that straight-line proximity is an invalid proxy for actual accessibility.

20. Unsupervised Machine Learning & Spatial Typologies (K-Means & PCA)

To uncover objective neighbourhood typologies, five standardized spatial features were extracted for each of the 344 grid cells: road density, intersection density, hospital distance, facility count, and accessibility score. Normalized via StandardScaler, K-Means was evaluated across k in [2, 7]. The Elbow Curve demonstrates diminishing inertia beyond k=5 (485.99), while the Silhouette Coefficient peaks at k=5 with a score of 0.392 (Figure 9).

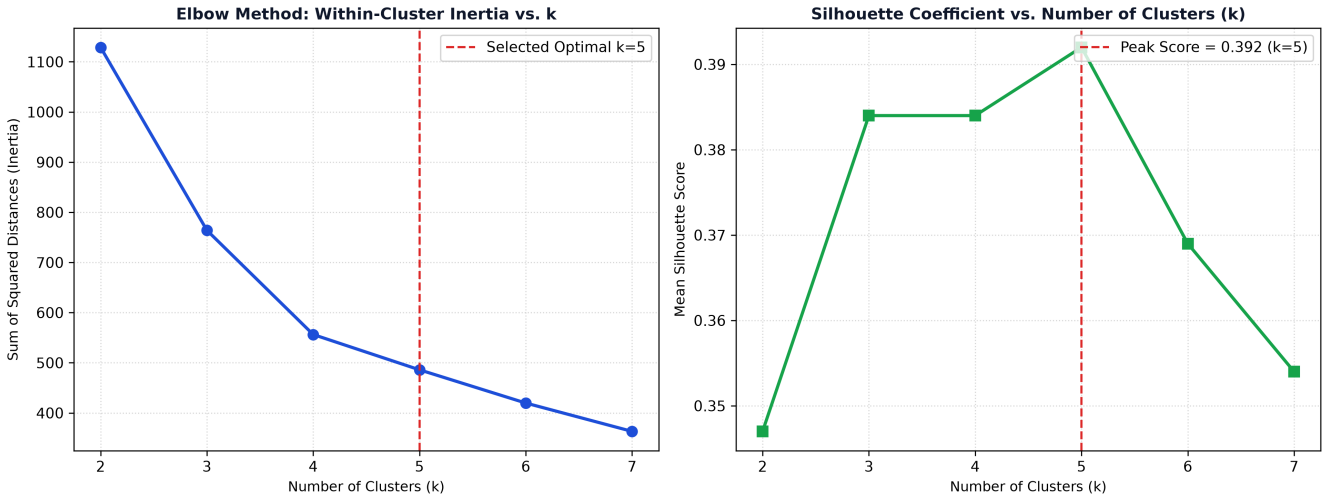


Figure 9: Unsupervised K-Means Clustering Evaluation (Elbow Inertia & Silhouette Curves).

2D Principal Component Analysis (Figure 10) captures 83.4% of total dataset variance (PC1: 55.1%, PC2: 28.3%). The spatial mapping of the 5 clusters (Figure 11 and Table 8) establishes distinct urban typologies: Cluster 0 (Dense Mixed Transit Sector, 129 cells), Cluster 1 (Hyper-Dense Urban Core, 102 cells), Cluster 2 (Peripheral Industrial Sector, 69 cells), Cluster 3 (Healthcare Anchor, 14 cells), and Cluster 4 (Commercial & Medical Buffer, 30 cells).

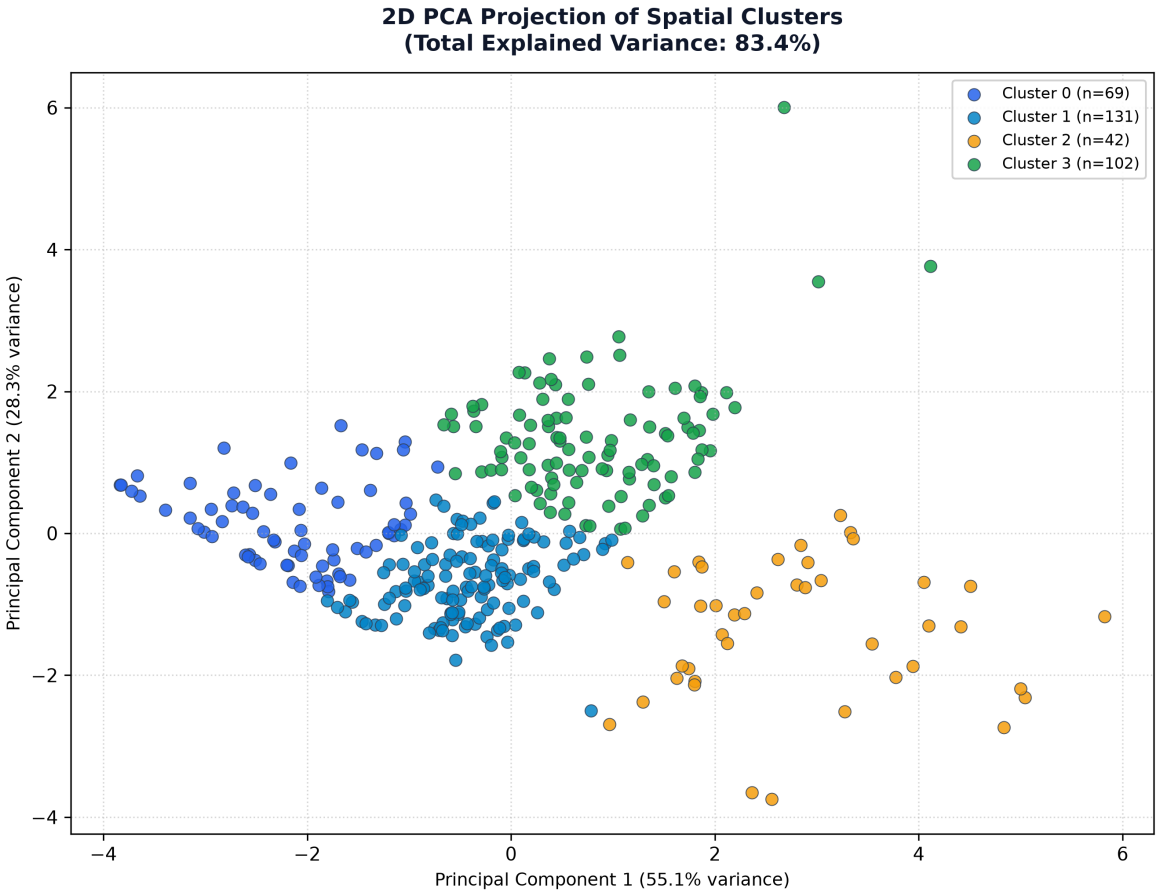


Figure 10: 2D Principal Component Analysis (PCA) Cluster Projection.

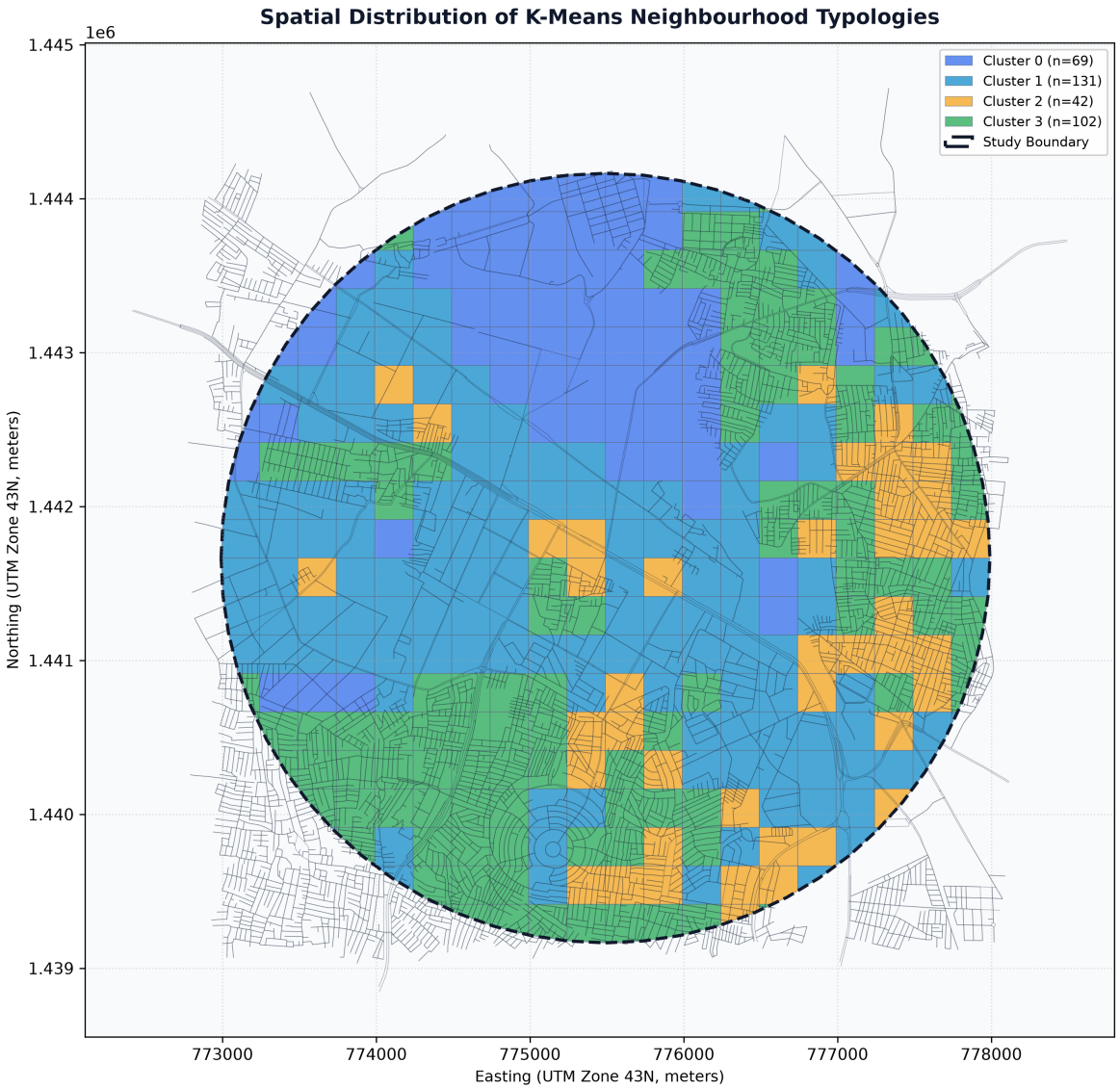


Figure 11: Spatial Distribution of Discovered K-Means Neighbourhood Typologies.

Cluster ID	Typology Designation	Cells	Mean Road Dens	Mean Hosp Dist	Access Score	Dominant Characteristics
Cluster 0	Dense Mixed Transit Sector	129	23.77 km/km²	416.4 m	70.9 / 100	Moderate road density, strong highway accessibility, solid hospital reach
Cluster 1	Hyper-Dense Urban Core	102	69.46 km/km²	545.1 m	70.7 / 100	Extremely dense residential street grid, high intersection density
Cluster 2	Peripheral Industrial Sector	69	14.34 km/km²	949.4 m	56.1 / 100	Low road density, railway yards, longest travel times to hospital care
Cluster 3	Healthcare Anchor Cluster	14	59.40 km/km²	125.8 m	97.5 / 100	Immediate walking proximity to major tertiary hospitals & high clinic count
Cluster 4	Commercial & Medical Buffer	30	48.08 km/km²	219.2 m	85.8 / 100	High amenity concentration along Tumkur Road & Dr. Rajkumar Road

Table 8: Unsupervised K-Means Cluster Characteristics & Sociodemographic Typologies.

21. Differentiation from Commercial Navigation Platforms (Google Maps)

Commercial mapping platforms (Google Maps, MapMyIndia) are engineered for individual navigation and commercial discovery. They do not provide structural graph analytics, continuous metric spatial tessellations, or unsupervised clustering. Table 9 articulates the fundamental distinctions between Smart Yeshwanthpur GIS, commercial platforms, and desktop GIS.

Analytical Dimension	Smart Yeshwanthpur GIS	Google Maps / Commercial Apps	Traditional Desktop GIS (QGIS)
Primary Objective	Urban infrastructure analytics & spatial equity	Consumer navigation & local business discovery	General-purpose standalone spatial modeling
Network Centrality Analysis	Explicit Betweenness & Degree computation	Unavailable (Proprietary black-box routing)	Manual processing via heavy desktop plugins
Physical Circuity Quant	Computed as d_network / d_haversine	Unavailable (hidden turn-by-turn routing)	Requires complex multi-step Python scripting
Grid Spatial Choropleths	Automated 250m metric grid scoring	Unavailable (no continuous spatial aggregation)	Manual rasterization or spatial join workflows
Machine Learning Typologies	Integrated K-Means & 2D PCA pipeline	Unavailable	Requires external scikit-learn Python scripts
Data Provenance & Openness	100% open-source OSM & transparent code	Proprietary, closed commercial data APIs	Open, but steep learning curve for non-specialists
Web Deployment & Usability	Zero-install Streamlit browser application	Consumer mobile/web client	Heavy local software installation required

Table 9: Multi-Dimensional Comparison: Smart Yeshwanthpur GIS vs. Google Maps vs. Traditional Desktop GIS.

22. Data Governance, Scientific Ethics & Analytical Disclaimers

1. **Topological Centrality vs. Live Traffic:** Betweenness centrality reflects graph structure. It identifies structural pinch points but does NOT measure live vehicular probe congestion.
2. **Spatial Accessibility vs. Clinical Quality:** The Accessibility Score measures physical distance and amenity presence, not medical quality, bed availability, or doctor specialization.
3. **Analytical Buffer vs. Official Wards:** The 2.5 km circular study buffer is an analytical convenience and does not represent official BBMP ward administrative boundaries.

23. Technical Limitations & Data Completeness

- **OSM Contributor Completeness:** Minor informal slum alleys and unregistered basement private clinics may be omitted from OSM databases.
- **Free-Flow Speed Model:** Edge travel times reflect statutory design speeds (15–50 km/h) rather than dynamic floating-car congestion delays.
- **Planar 2D Geometry:** Flyover ramps are represented without full 3D grade-separation elevation profiles.
- **Unsupervised ML Bounds:** K-Means clusters identify empirical mathematical patterns and do not imply predictive causality.

24. Future Roadmap & Proposed Enhancements

1. **Real-Time Transit Feeds:** Ingesting live BMTC and Namma Metro GTFS-RT feeds for multi-modal public transit isochrones.
2. **Dynamic Probe Congestion:** Incorporating TomTom or Uber Movement open speed APIs for time-of-day impedance routing.
3. **Two-Step Floating Catchment Area (2SFCA):** Implementing 2SFCA accessibility incorporating verified hospital bed capacities against ward census demand.
4. **City-Wide Scaling:** Expanding the automated pipeline across all 198 BBMP wards in Greater Bengaluru (712 sq km).

25. Practical Policy Implications & Urban Planning Diagnostics

- **Targeted UPHC Placement:** Establish 24/7 Primary Urban Health Centers in Grid Cells GRID_0151 and GRID_0131 where hospital driving distances exceed 2.0 km.
- **Railway Yard Permeability Underpasses:** Construct light-vehicle and pedestrian underpasses beneath the railway yard connecting Subhash Nagar to Dr. Rajkumar Road to collapse the 3.58x circuity factor.
- **Bottleneck Corridor Protection:** Enforce strict parking bans and deploy Intelligent Traffic Signals along the 794 high-betweenness segments on Tumkur Road approaches.
- **Feeder Transit Buffers:** Deploy electric feeder mini-buses linking Mattikere residential pockets to Yeshwanthpur Metro interchange.

26. Technical Conclusion

Smart Yeshwanthpur GIS establishes an open-source, mathematically rigorous spatial analytics platform tailored to the complex infrastructure realities of urban India. By moving beyond naive straight-line assumptions and proprietary commercial tools, the project equips municipal authorities with empirical insights into road network vulnerability and healthcare equity. The system models 15,798 road segments, isolates 794 structural bottlenecks, evaluates 84 healthcare amenities across 344 metric grid cells, quantifies severe circuity detours imposed by the railway yard, and uncovers five distinct neighbourhood typologies through unsupervised machine learning. Deployed as an interactive Streamlit application with complete GIS vector data exports, the platform stands as a robust, extensible prototype for data-driven smart city governance in Bengaluru and expanding Global South metropolises.

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